

Module 01: Systems in Crochet Circles

Learning Objectives

1. Identify the boundary between your crochet work and its environment using the concept of a **Markov blanket**.
2. Recognize how hook, yarn, and hands form an interacting system with internal and external states.
3. Map the components of a crochet project onto the formal structure of a system in Active Inference.

Introduction

Pick up a crochet hook and a ball of yarn. Before you make a single stitch, you already have a system: the hook in your hand, the yarn trailing from the skein, and the space where fabric will soon appear. In Active Inference, a **system** is anything that maintains itself as distinct from its surroundings. Your crochet work-in-progress does exactly this — it has an inside (the stitches already made), an outside (the yarn not yet worked, the air around the fabric, the table beneath it), and a boundary where the two meet.

This module invites you to look at your crochet practice as a living example of systems thinking. Where does the project end and the world begin? What role does the edge of your fabric play? And how does the simple act of hooking yarn through a loop create and maintain a boundary that defines the whole work?

Key Concepts

1. The Hook-Yarn Boundary as Markov Blanket

In Active Inference, a **Markov blanket** is the statistical boundary that separates an agent's internal states from the external world. It consists of **sensory states** (how the outside world affects the inside) and **active states** (how the inside affects the outside).

In crochet, this boundary is tangible. The **edge of your fabric** — the live loops sitting on your hook, the working row you are currently stitching — is exactly the Markov blanket of your project. Everything behind the working row is *internal state*: the finished rows, the structure you have built, the pattern that has taken shape. Everything ahead of the working row is *external state*: the unworked yarn, the pattern instructions you haven't reached yet, the eventual shape the project will take.

The live loop on your hook is special. It is the point of contact, the sensory-active boundary. Through it, you **sense** the tension of the yarn (sensory state) and you **act** on the fabric by pulling new loops through (active state). If you set your work down and the live loop slips off, the boundary is compromised — the fabric can unravel. The Markov blanket dissolves.

This is not just a metaphor. In the formal mathematics of Active Inference, the Markov blanket is defined by conditional independence: internal states and external states only communicate through blanket states. In your crochet, the finished rows (internal) and the ball of yarn (external) only interact through the working edge — the hook, the live loop, and your hands.

2. Internal States: The Work-in-Progress

The **internal states** of a system are everything that is shielded from the environment by the Markov blanket. In crochet, this is the fabric you have already created: the foundation chain, the completed rows, the emerging pattern of single crochets, double crochets, or whatever stitches you have chosen.

These internal states carry information. The stitch count tells you where you are. The height of the rows tells you how far along you are. The texture and drape of the fabric tell you whether your tension has been consistent. In Active Inference terms, internal states encode the system's **beliefs** about itself — they are the accumulated result of all the actions (stitches) the system has taken.

Notice that once a row is completed and you have turned your work, those stitches become relatively fixed. They are no longer directly in contact with the external world. They influence the current row (because you crochet into them), but they do so through the boundary — through the top loops of the previous row, which are the points of insertion for your hook. This

layered structure, where past actions become the substrate for future actions, is a hallmark of how internal states function in Active Inference.

3. External States: Yarn Supply and Environment

The **external states** are everything outside the Markov blanket. In crochet, this includes the yarn still in the skein, the pattern you are following, the lighting in the room, the other members of your crochet circle, and even the intended recipient of the finished project.

The yarn supply is the most immediate external state. It feeds into the system through the working yarn — the strand that runs from the skein to your hook. The crocheter does not directly perceive the entire ball of yarn at once; you feel it through the tension on the strand, the weight of the skein on your lap, the way the yarn feeds smoothly or catches.

The pattern instructions are another kind of external state. They exist outside your project, but they influence what you do at the boundary. When you read "ch 3, dc in next st," you are receiving information from the external world that shapes your actions at the Markov blanket.

This is why crochet in a group is so rich: the external states expand to include the projects of others, their advice, their shared patterns. The boundary of your system is clear — your hook, your yarn, your fabric — but it exists within a much larger environment.

Applications

In crochet, we see Systems manifest in:

- **Gauge Swatches as System Calibration:** When you crochet a gauge swatch before starting a project, you are essentially calibrating your system. You establish how your internal states (tension, hook size, stitch height) interact with external states (yarn weight, pattern requirements) at the boundary. The swatch lets you check whether the Markov blanket of your system — the edge where stitches form — is producing the expected results.
- **Frogging as System Boundary Repair:** When you discover an error and rip back your work ("frogging," from "rip it, rip it"), you are deliberately dissolving internal states back through the Markov blanket. Completed stitches return to unworked yarn —

internal states become external states again. This is a dramatic illustration of how the system boundary is maintained by active work, and how it can be reversed when the internal states no longer match expectations.

Conclusion

Every crochet project is a system in the Active Inference sense: it has internal states (the fabric you have made), external states (the yarn and world around it), and a Markov blanket (the live edge where hook meets yarn). Understanding this structure helps us see why the working row feels so alive, why a dropped stitch is so alarming, and why the simple act of pulling yarn through a loop is an act of system maintenance. In the next module, we turn our attention to the agent at the center of this system: you, the crocheter.